

Anti-insecticides activity in cell-lines model of *Thunbergia laurifolia* leaf extract aiming for functional drink

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Abstract

Thunbergia laurifolia has been used for folklore medicine and consumed as an herbal tea in Thailand ancient times. The aim of this study was to determine the influence of aqueous crude extracts of *T. laurifolia* leaves on endogenous antioxidant enzyme activity, lipid peroxidation indicated by malondialdehyde (MDA) production, and anti-insecticides property using three types of cells. Murine macrophage (RAW264.7), human embryonic kidney (HEK293), and human hepatocellular carcinoma (HepG2) cells, were treated with crude extract either combined, pre, and post-treatment exposure to chlorpyrifos (CP) and methomyl (MT). Cell viability was determined using MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) test to find out anti-insecticide properties. Results indicated that crude extract increased catalase (CAT) and glutathione peroxidase (GPx) activities while decreasing malondialdehyde (MDA) levels in HEK293 and HepG2 cells and improved cells viability in all cell types induced by CP and MT toxins. The crude extract provided a protection and recovery effect on CP and MT insecticides better than atropine sulfate (AS) standard drug especially in HepG2 and HEK293 cells. Therefore, it was concluded that crude extract can improve antioxidant enzymes level and can be used as anti-insecticides. However, more investigation is needed to play it safe for functional drinks by processing in animal model further.

Keywords: *Thunbergia laurifolia*; antioxidant enzymes; anti-insecticides; cell-lines.

Practical Application: *Thunbergia laurifolia*, an herbal tea in Thailand, is a source of antioxidants and bioactive compounds. It is important when seeking to increase the levels of antioxidant enzyme and for protection against insecticides.

1 Introduction

It is known that Thailand has been accepted as the kitchen of the world. However, pesticide usage in crop protection to increase the production levels, quality, and appearance is still under consideration (Panuwet et al., 2012). As humans are at the top of the food chain, food containing toxic substances can potentially accumulate in the human body (Sapbamrer et al., 2011). Organophosphates such as chlorpyrifos (CP) and carbamate insecticide including methomyl (MT) are very toxic and hazardous insecticides (Tamimi et al., 2006; Li et al., 2015), and they are banned in many countries. Their toxicity has been reported to relate to acetylcholinesterase inhibition, one of the important enzymes involved in the human nervous system (Amritha & Kaliwal, 2016).

The market of functional foods in many countries exhibited rapid growth in the present-day such as using probiotics in dairy products and bioactive compounds from plants (Gaspar-Pintilieșcu et al., 2020; Granato et al., 2020; Balthazar et al., 2021). Plants have important sources of bioactive compounds that can function on antioxidant, anti-inflammatory, antimicrobial, anti-wrinkle, anti-proliferative activity, and hypercholesterolemia controlling (Rafiq et al., 2020; Aslam et al., 2021; Barbosa et al., 2021; Byun et al., 2021; Eor et al., 2021; Iwansyah et al., 2021).

T. laurifolia, with the Thai name 'Rang Jued', belonging to the family Acanthaceae, (Morkmek et al., 2010) has been used in Thai traditional medicine for many centuries. Extracts of its fresh and dried leaves, barks, and roots are mainly used as an antidote for treating insecticides, drugs, heavy metals, alcohol, and food poisoning including chemical toxins (Thongsaard & Marsden, 2002; Oonsivilai, 2006; Inta et al., 2013; Rocejanasaroj et al., 2014; Maneenoon et al., 2015). Moreover, this plant has been stated to have antioxidant and anti-inflammatory properties (Chan et al., 2012; Wonkchalee et al., 2012). Previous studies have demonstrated that the aqueous crude extract of *T. laurifolia* leaves contains a high number of phenolics and flavonoids, and provides a protective effect against Cd-induced oxidative stress in the cell line (Junsi et al., 2017a; Junsi et al., 2017b; Junsi et al., 2020). It is known that the liver and kidney organs are the detoxification organs in humans while macrophages are the immune cell type that responds as part of the first defense against pathogens (Helali et al., 2016). Therefore, the objective of this study was to determine the effect of the crude extract of *T. laurifolia* on endogenous antioxidant enzyme activities (CAT and GPx) and MDA production, in addition, to determining the protection and healing using RAW264.7, HEK293 and HepG2 cells as test models affected by CP and MT insecticides, which are commonly used in fresh products.

Received 07 Jan., 2022

Accepted 22 Feb., 2022

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2 Materials and methods

2.1 Chemicals and reagents

The chemicals used for the determination of the cell culture were purchased from Gibco (Carlsbad, California). Most of the chemicals used for the determination of the antioxidant enzymes and anti-insecticides were purchased from Sigma-Aldrich (Seelze, Germany), otherwise they have been specified later. The solvents and all chemicals and reagents were special-grade reagents.

2.2 Sample preparation and extraction

The *T. laurifolia* leaves in the developing stage which purchased from the contact farmer (Bangkok, Thailand). They were washed with tap water, drained and air-dried for 5-8 days following the folk medicine method to obtain a moisture content of 8-10% (w/w). It was then ground to a fine powder using a 20-40 mesh. The powder of the leaves was then soaked in hot water, $98 \pm 1^\circ\text{C}$ (1 : 10 w/v) for 1 h, and filtered through three layers of gauze followed by Whatman No. 4 filter paper. The filtrate was freeze-dried and stored at 4°C for further study as a dried crude extract (Ruangyuttikarn et al., 2013).

2.3 Cell culture

The murine macrophage RAW264.7 cells and human embryonic kidney cells (HEK293) were purchased from the American Type Culture Collection (ATCC, USA). They were all grown in RPMI-1640 medium and minimum essential medium (MEM) respectively, while the human hepatocellular carcinoma cells (HepG2) was kindly provided by Assoc. Prof. Dr. Teerapol Srichana (Faculty of Pharmaceutical Sciences, Prince of Songkla University, Thailand) cultured in Dulbecco's Minimal Essential Media (DMEM). The media were supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin. The cells were cultured at 37°C in a humidified atmosphere of 5% CO_2 and 95% air in a fully humidified incubator. The cells were harvested with 0.25% trypsin-EDTA and suspended in a fresh medium (American Type Culture Collection, 2014).

2.4 Determination of the crude extract on antioxidant enzyme activity and malondialdehyde value

Preparation of the endogenous cellular extract

All cell lines were seeded at 1×10^6 cells/mL in 60 mm tissue culture dishes and incubated for 24 h to allow the cells to grow and adhere to the dish. In each culture dish, the cells were washed twice with 2 mL of PBS (pH 7.2) before being treating with the crude extract (10-1,000 $\mu\text{g/mL}$). The crude extract was added to the dish and incubated for 24 h at 37°C before its removal from the cell cultures. The cells in each culture dish were harvested by incubation with 0.5 mL of 0.25% trypsin-EDTA. Then, 1 mL of culture media was added and centrifuged at $1,000 \times g$ for 10 min. The cell pellets were washed with ice-chilled PBS (500 mL, 2 times) and then lysed by sonication on ice for 1 min using a probe-type sonicator (Vibra-Cell, Sonics and Materials Inc., Newtown, CT, USA) following the method of Du et al. (2016).

The mixture was then centrifuged at $10,000 \times g$ for 10 min at 4°C and the supernatant (endogenous cellular extract) was assayed for enzyme activity and malondialdehyde value.

Catalase (CAT) activity, Glutathione Peroxidase (GPx) activity, and Malondialdehyde (MDA) value

The activity of CAT was determined by monitoring the decrease in absorbance at 240 nm due to H_2O_2 consumption following the method of Aebi (1984) with minor modifications. The activity of GPx was determined based on glutathione oxidation by GPx in the presence of DTNB (Flohé & Günzler, 1984). To measure the MDA value, the modified method of Heath & Packer (1968) was used. We measured the absorbance at 532 nm using MDA as an external standard. Both enzyme activities and MDA value were calculated as the enzyme unite and MDA μmol per mg protein according to the calibration curve of the bovine serum albumin (BSA) standard using Bradford's assay (Bradford, 1976) and expressed as a percentage of the control.

2.5 Determination of the crude extract's anti-insecticide properties

The cytotoxicity of both chlorpyrifos (CP), a representative of organophosphate and methomyl (MT) methyl carbamate insecticides, was tested on RAW264.7, HEK293, and HepG2 cells using the MTT assay Mosmann (1983). The percentage viability of the population was calculated using the equation (Equation 1):

$$\% \text{ Cell viability} = \left[\frac{\text{Absorbance of sample}}{\text{Absorbance of control}} \right] \times 100 \quad (1)$$

The effect of the crude extract on each cell type induced by each insecticide at CC_{50} was investigated. In this experiment, atropine sulfate (AS) was used as a positive control. Briefly, the cells were plated in a 96-well plate at a density of 1×10^6 cells/mL and we allowed the cells to attach to the culture plate (2 h for RAW264.7 and 24 h for HEK293 and HepG2 cells). The testing group was treated with insecticide at CC_{50} and divided into three groups as the combined-treatment, pre-treatment, and post-treatment. The crude extract was allowed to have contact time with the cells for 24 h as follows:

- Group 1: (Combined-treatment) the crude extract or AS was added to the insecticides at CC_{50} together with the cells and incubated for 24 h.
- Group 2: (Pre-treatment) the crude extract or AS was added to the cells and incubated for 24 h before adding the insecticides (CC_{50}) and being incubated for 24 h.
- Group 3: (Post-treatment) the insecticides (CC_{50}) was added to the cells and was incubated for 24 h before adding the crude extract or AS and incubated for 24 h.

For the normal control, the media was added to the cells without the crude extract, AS and insecticides (CC_{50}). For the negative control, each insecticide (CC_{50}) was added to the cells without the crude extract or AS. The amount of cell viability was detected using a MTT cytotoxicity assay Mosmann (1983) and the percentage of cell viability was calculated using Equation 1.

2.6 Statistical analysis

A completely randomized design (CRD) was used throughout this experiment. The data were subjected to analysis of variance (ANOVA). Comparison of the means was carried out using Duncan's multiple range tests. Significance was declared at $p < 0.05$ using the statistical software.

3 Results and discussion

3.1 Antioxidant enzyme activity and malondialdehyde value

The effect of the crude extract on the antioxidant enzyme activities (CAT and GPx) including the MDA level in all cell types, and RAW264.7, HEK293, and HepG2 cells is shown in Figure 1. CAT is the first antioxidant enzyme to convert hydrogen peroxide into water and oxygen, and its catalytic reaction is a one-step process. GPx is the main scavenger of hydrogen peroxide by catalyzing the oxidation of glutathione and using hydroperoxides as substrates (Magalingam et al., 2013). The result show that the activity of CAT and GPx in the HEK293 and HepG2 cells were higher compared with the control but not present in the

RAW264.7 cell. The increase of both enzymes in HEK293 and HepG2 has been reported and discussed in our previous work (Junsi et al., 2020) which may be via the phenolic compounds and essential elements (Fe, Zn, Cu, and Se). These are co-factors of the antioxidant enzymes CAT and GPx in the crude extract. The increment of CAT and GPx enzymes did not appear in the RAW264.7 cells (Figures 1A-1B) when the crude extract content increased. Moreover, the activity of both antioxidant enzymes from the RAW264.7 cell decreased when the concentration of crude extract increased was compared with the control. There have been several scientific studies that have reported that phenolic and flavonoid compounds, as well as trace elements, can act as pro-oxidant and induce oxidative stressors in the cells (Galati & O'Brien, 2004; Halliwell, 2008; Kyselova, 2011; Halliwell, 2014; Nanda & Agrawal, 2016). Therefore, it has been pointed out that the higher cytotoxicity property of the crude extract tested in the RAW264.7 cell is related to the higher extract concentration similar to our previous study (Junsi et al., 2017a). However, the HEK293 and HepG2 cells did not exhibit the same phenomenon maybe due to the strong antioxidant enzymes activity. Both liver and kidney cells contain a higher level of antioxidant enzyme

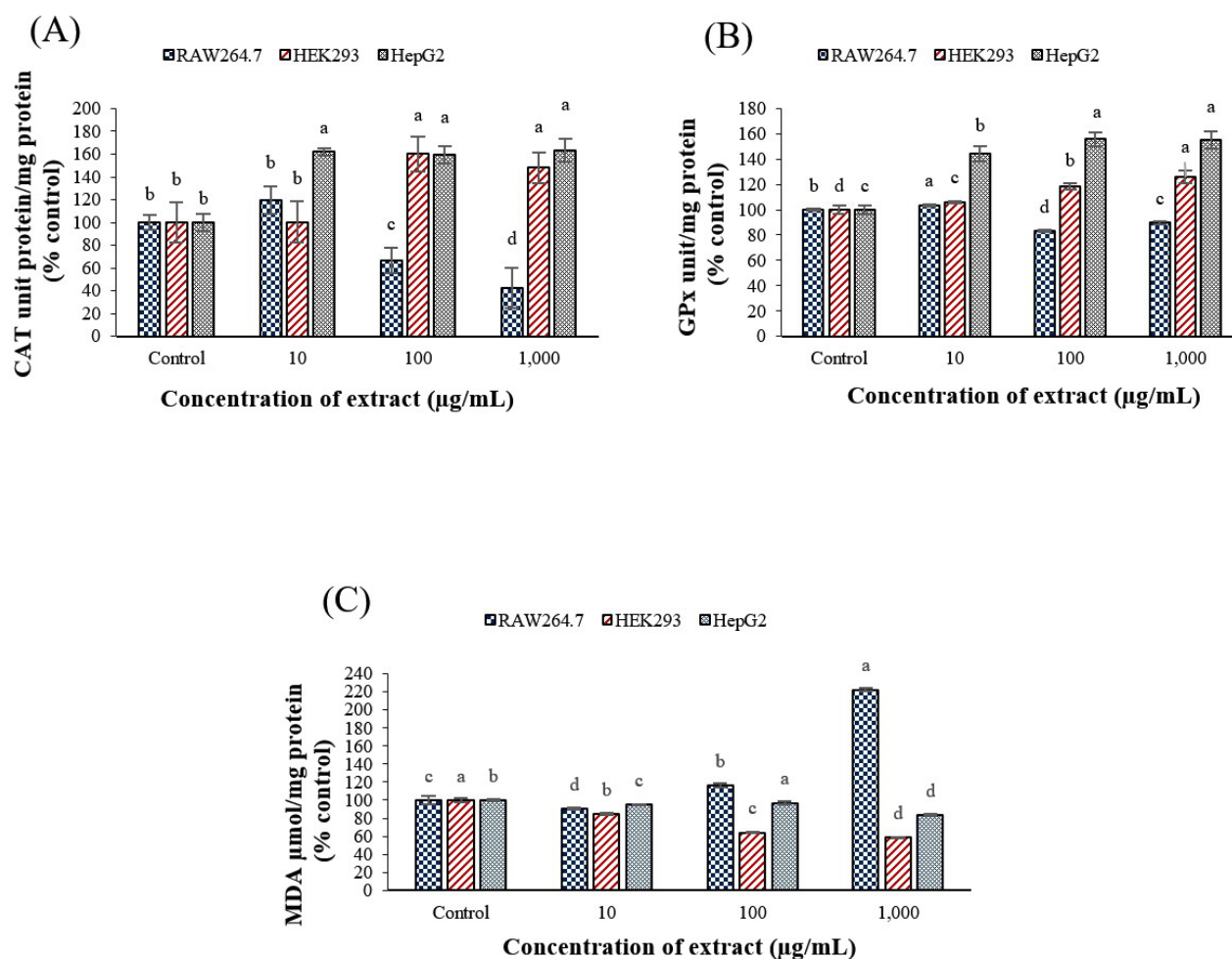


Figure 1. Antioxidant enzymes activities and malondialdehyde value of endogenous cellular extract treated with *T. laurifolia* extract in RAW264.7, HEK293 and HepG2 cells; (A) catalase; (B) glutathione peroxidase and (C) malondialdehyde. ^{a-d} mean within cell types with different letters are significantly different ($p < 0.05$). Values are represented as the mean $\pm$ standard deviation ($n = 3$).

activity (Schriner & Linford, 2006) which may help to better balance the homeostasis of the cell following oxidative stress compared to other cells.

The level of MDA in this experiment is shown in Figure 1C. Typically, MDA is used as the end product of lipid peroxidation which is the widely accepted concept of cell damage leading to various diseases (Magalingam et al., 2013). The results show that MDA level significantly decreased in the HEK293 and HepG2 cells while it increased in the RAW264.7 cells when the concentration of the crude extract increased, related to the increase in the CAT and GPx enzyme activities. There is some scientific data that shows that both CAT and GPx enzymes are involved with hydrogen peroxide elimination which is directly related to MDA reduction (Harman, 1991; Jung et al., 2006). The reduction of the MDA level, when treated with the crude extract, may indicate oxidative stress protection. Abdallah et al. (2012) reported that the combination of caffeic acid and quercetin pretreatments significantly reduced the levels of MDA from Lambda-cyhalothrin, an insecticide toxin inducing oxidative stress in rat erythrocytes. However, from the results of the antioxidant enzymes and MDA level of the crude extract on each cell type, it has been pointed out that use of this crude extract may include needing to pay more attention to even healthy people who should not drink it too much. It is better for it to be further studied in both animal and clinical trials to ensure its safety.

3.2 Cytotoxicity of CP and MT

Chlorpyrifos (CP), an insecticide that originated from an organophosphate, was treated in each cell type and the CC_{50} value was reported (Table 1). It was found that the CP insecticide could induce cell death from the highest to lowest concentration as followed; RAW264.7 > HepG2 > HEK293 respectively. The results indicate that CP had the ability to induce HEK293, a kidney cell death with the lowest concentration compared with the other cells. The results also show that the RAW264.7 cells were the most resistant to CP. This effect may due to the composition of the cell membrane and the polarity of CP. Schoeniger et al. (2011) reported that the RAW264.7 macrophage cells contained a high lipid composition in the membrane while CP has a non-polar or less water solubility (World Health Organization, 2004). As it is known that the cell culture system with a high polarity does not facilitate the CP toxin getting through the cells with a fat content, the survival rate high of RAW264.7 was significantly noticed. Moreover, in this result, HepG2 resisted CP more than HEK293. This is similar to the findings of Xing et al. (2013) who reported that CP could decrease the acetylcholinesterase (AChE) activity in the kidneys more than the same in the liver of common carp (*Cyprinus carpio*). The results of cytotoxicity at CC_{50} of MT, a carbamate insecticide, showed that MT significantly

induced cell death in RAW264.7 compared with HEK293 and HepG2. This result was not in agreement with the CC_{50} of CP which indicated the least induced toxin in the RAW264.7 cells. This may be because MT has more of a polarity compared to CP, therefore it can dissolve in the system and easily make contact with the cells, leading to inducing toxicity more than CP. The CC_{50} of CP and MT was then selected as the working concentration for assessing the anti-CP and anti-MT toxicity property of the crude extract and AS respectively.

3.3 Anti-CP property of the crude extract and AS on cells

The effect of the crude extract on CP's toxicity property on the RAW264.7, HEK293, and HepG2 cells is shown in Figure 2. It was found that the crude extract could increase HepG2 cell survival significantly which then decreased at higher concentration of extract when the cell was treated with CP in all treatments, including combined, pre-, and post-treatment. The crude extract in the combined treatment can improve RAW264.7 cell survival (Figure 2A) at 10 to 500 $\mu\text{g/mL}$ of extract. CP toxicity was proposed to cause cell death via oxidative stress (Verma & Srivastava, 2003; McCarthy et al., 2004) through the cell membrane composed of poly-unsaturated fatty acids, a primary target for reactive oxygen attacks leading to cell membrane damage (Repetto et al., 2012). Therefore, it has been pointed out that the crude extract contains a high amount of phenolics with a high number of antioxidants. This response leads to an oxidative stress reduction leading to the high cell viability found in this experiment. This is similar to the findings of Usanawarong et al. (2000) who showed that the aqueous leaf extract of *T. laurifolia* can decrease plasma malonaldehyde (MDA), an indicator of lipid peroxidation-derived from the free-radical mediated reactions of a rat with paraquat. Moreover, Lertpongpipat & Chaiyakhun (2011) reported that drinking *T. laurifolia* tea could increase the level of cholinesterase, an enzyme located in the intercellular space that is responsible for acetylcholine degradation in the human nervous system. This important enzyme was inhibited by organophosphate and carbamate insecticides affecting the degradation of acetylcholine and unbalancing the equilibrium leading to a variety of short-term and chronic effects such as a headache, lowered heartbeat, and visual disorders including cancer (Eleršek & Filipič, 2011). However, in this experiment, the results revealed that the crude extract could not increase cell survival in HEK293. Moreover, a high concentration of crude extract induced more cell death. It was pointed out that the biological activity of any active compound may depend on many variable factors such as the quantity and quality of bioactive compounds as well as the type and functionality of the cellular target. Furthermore, the results show that the CP was more toxic to the HEK293 and HepG2 cells than the RAW264.7 cells which is related to the cytotoxicity property of CP on cells (Table 1). The cell viability of RAW264.7 cells in the negative control (cell only contact with CP) was due to its better resistance and recovery compared to the other cell types (Figures 2B-2C).

AS was used as the positive control in this experiment. The results indicate that AS can decrease RAW264.7 cell death in the combined treatment (Figure 3) but with less function compared to the other cells and other treatments. It has been

Table 1. 50% cytotoxicity concentration (CC_{50}) of CP and MT toxicity on cells viability determined by MTT assay.

Cell types	CC_{50} of CP ($\mu\text{L/mL}$)	CC_{50} of MT (mg/mL)
RAW264.7	0.25	0.54
HEK293	0.11	3.88
HepG2	0.13	5.78

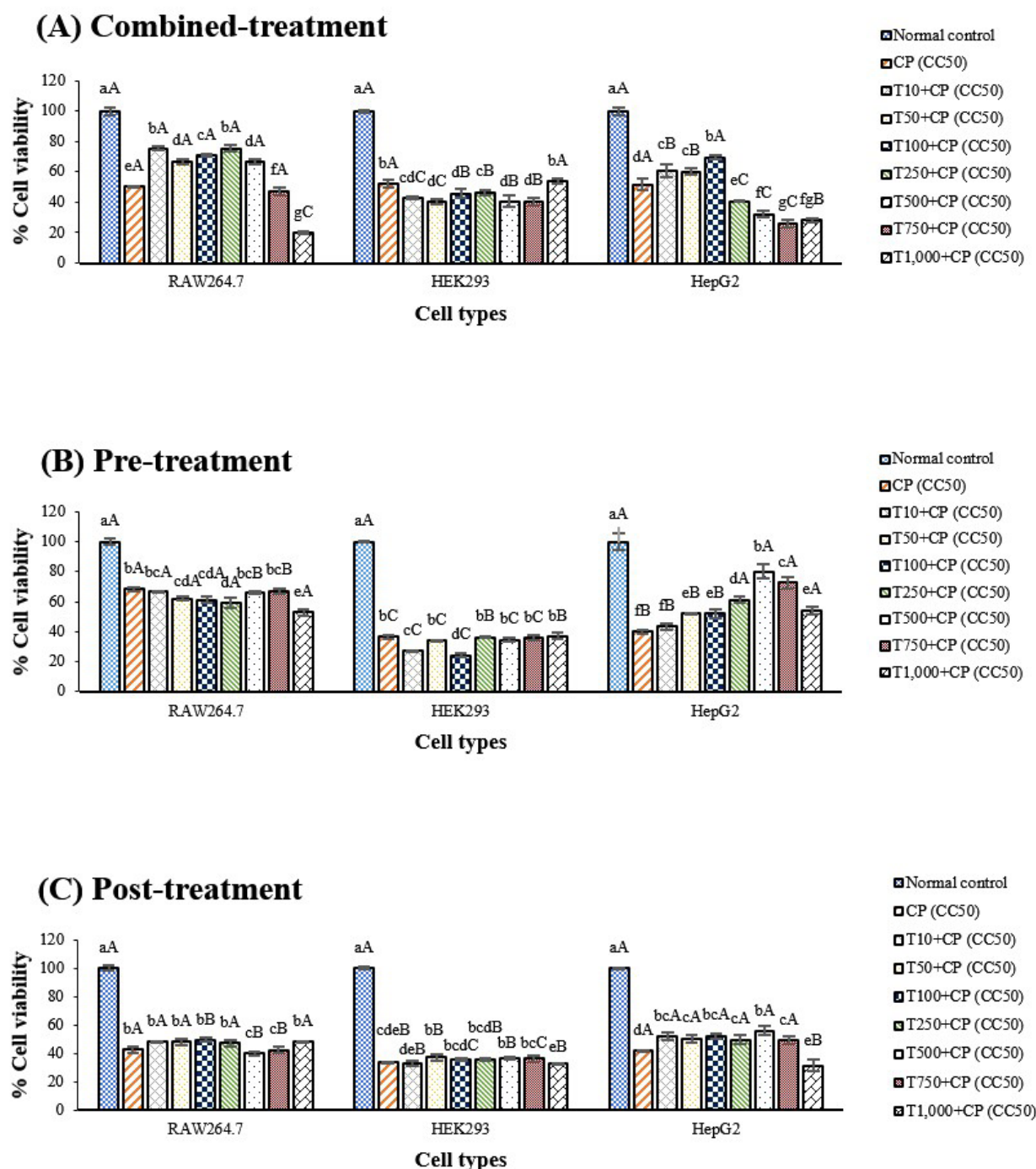


Figure 2. Anti-CP toxicity of the aqueous crude extract of *T. laurifolia* leaves in RAW264.7, HEK293 and HepG2 cells determined by MTT assay; (A) combined-treatments; (B) pre-treatment and (C) post-treatment with crude extract. ^{a-g} mean within cell types with different letters are significantly different ($p < 0.05$); ^{A-C} mean within an extract concentration with different letters are significantly different ($p < 0.05$); T means the crude extract of *T. laurifolia* leaf ($\mu\text{g/mL}$). Values are mean $\pm$ standard deviation ($n = 3$).

pointed out that AS can protect against cell death due to CP especially in the combined treatment, where AS and CP were added together in the cell. As is well known, CP poisoning can be divided according to the site of acetylcholine accumulation in the organism; the central neural system, peripheral autonomic nerve system, and nicotine receptors, as well as the muscarine

receptors found in the kidney, liver, and macrophage cells (Li et al., 2009; Eleršek & Filipič, 2011; Hussmann et al., 2011; Koarai et al., 2012). The mechanism of AS involves the inhibition of muscarinic acetylcholine receptors, which is the cause of muscarinic cholinergic toxidromes such as indigestion, a lowered heartbeat, and visual disorders (Eddleston & Chowdhury, 2016).

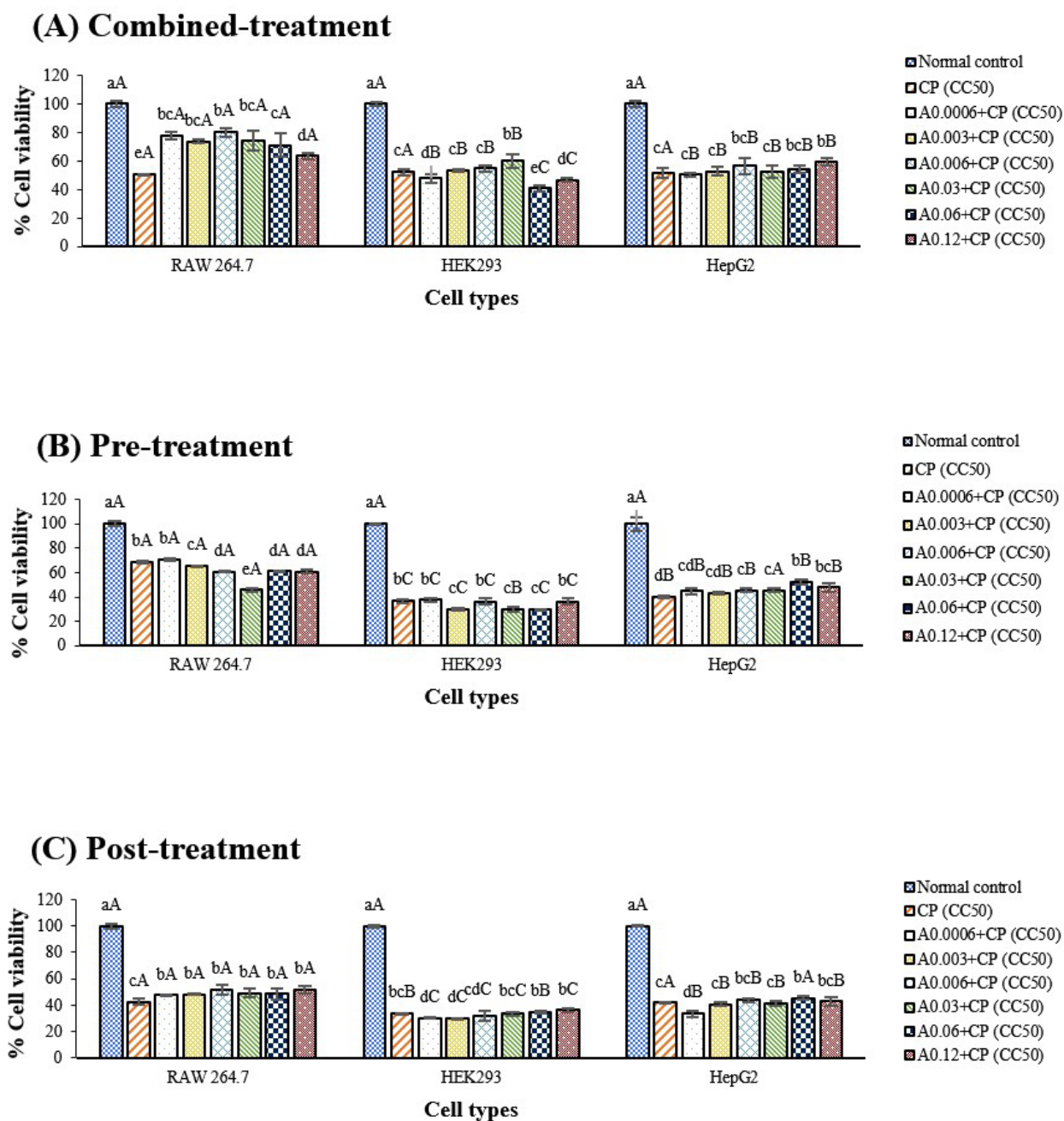


Figure 3. Anti-CP toxicity of AS in RAW264.7, HEK293 and HepG2 cells determined by MTT assay; (A) combined-treatments; (B) pre-treatment and (C) post-treatment with crude extract. ^{a-c} mean within cell types with different letters are significantly different ($p < 0.05$); ^{A-C} mean within an AS concentration with different letters are significantly different ($p < 0.05$); A means atropine sulfate solution ($\mu\text{g/mL}$). Values are mean $\pm$ standard deviation ($n = 3$).

However, AS has a short half-life (3.0 ± 0.9 h in an adult) and it is easily destroyed by enzymatic hydrolysis, particularly in the liver (DrugBank Online, 2018). This may not help in the pre-treatment (cell contact with AS 24 h before treated with CP). The results indicate that the crude extract provided an anti-CP property better than the AS standard drug in both the pre and post-treatments. Therefore, the crude extract might have both a protective and healing effect even better when compared with

the AS standard drug. This may be due to the multiple functions of the various compounds including phenolics, flavonoids and other material compounds which support cell growth and inhibit the oxidation process.

3.4 Anti-MT property of the crude extract and AS on cells

The effect of the anti-MT toxicity property of the crude extract was tested in similar ways to the CP toxicity assay, and

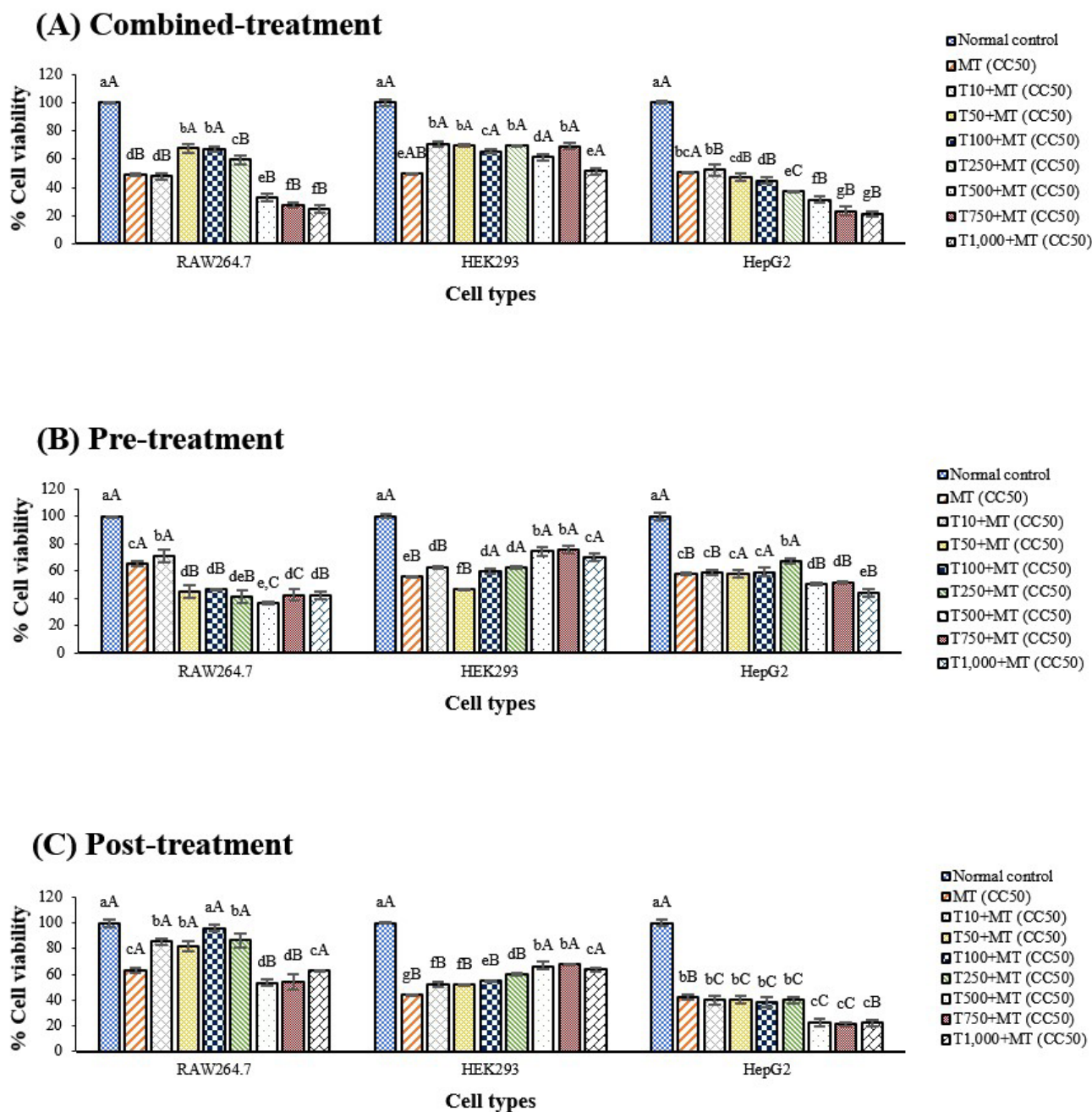


Figure 4. Anti-MT toxicity of the aqueous crude extract of *T. laurifolia* leaves in RAW264.7, HEK293 and HepG2 cells determined by MTT assay; (A) combined-treatment; (B) pre-treatment and (C) post-treatment with crude extract. ^{a-g} mean within cell types with different letters are significantly different ($p < 0.05$); ^{A-C} mean within an extract concentration with different letters are significantly different ($p < 0.05$); T means the crude extract of *T. laurifolia* leaf ($\mu\text{g/mL}$). Values are mean $\pm$ standard deviation ($n = 3$).

the results are shown in Figure 4. It was revealed that the crude extract could help the HEK293 cells survive in all treatments. In addition, the crude extract showed an increase in cell viability in the combined and post-treatments for the RAW264.7 cells. However, the crude extract seemed not to help at higher concentrations, and the crude extract even induced cell death in the HepG2 cells. It is implied that higher cytotoxicity could be found in some cell types due to the nature of the cell and

dose which differed from one to another. Moreover, the cell viability in RAW264.7 clearly decreased when the concentration of the extract exceeded 500 $\mu\text{g/mL}$ in both the combined and post-treatment groups (Figures 4A-4C). This phenomenon is in agreement with the cytotoxicity activity of the extract that was reported in a previous study where it showed there to be a dose-dependent reduction of cell viability when the concentration of the crude extract increased (Junsi et al., 2017a). It is also related

to the results of the CAT and GPx decreasing and increasing regarding the MDA level in the RAW264.7 cells. Therefore, it is implied that both phenolic and flavonoid compounds can have antioxidant and pro-oxidant properties depending on many factors including dosage, the structure of the polyphenol, the cellular redox status, and the presence of metal ions in the system (Yordi et al., 2012; Kim et al., 2014) as well as the cell target. Moreover, some phenolic and flavonoid compounds such as gallic acid and epigallocatechin gallate (EGCG) were reported to induce H_2O_2 generation, which is a cause of oxidative stress either intracellular and/or in cell culture medium, and it is toxic to cells (Halliwell, 2008, 2014). Some flavonoids, such as apigenin,

quercetin, naringenin, and myricetin also induced apoptosis via a pro-oxidant action (Galati & O'Brien, 2004; Kyselova, 2011).

The result of the AS (positive control) for both anti-CP and MT properties provided greater protection in terms of cell viability (Figure 3, Figure 5) especially in the RAW264.7 cells. Generally, RAW264.7 consists of macrophage cells that are reported to contain the most contain choline acetyltransferase, choline transporters, AChE, and acetylcholine (Kawashima & Fujii, 2004; Razani-Boroujerdi et al., 2008). Therefore, any component function that inhibits the system involving acetylcholine which is related to AS action should have more of a protective effect in the RAW264.7 cells.

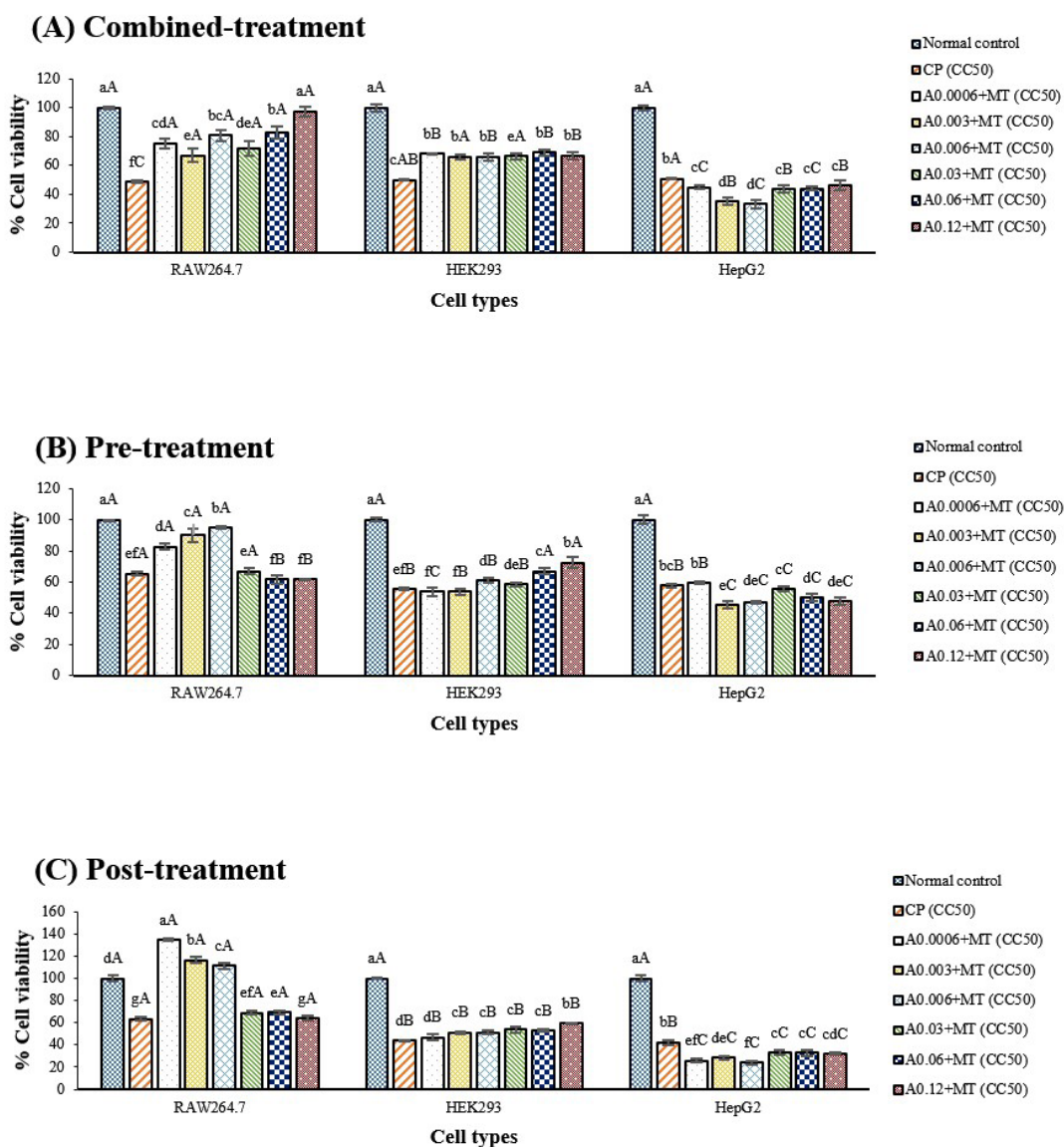


Figure 5. Anti-MT toxicity of AS in RAW264.7, HEK293 and HepG2 cells determined by MTT assay; (A) combined-treatments; (B) pre-treatment and (C) post-treatment with crude extract. ^{a-f} mean within cell types with different letters are significantly different ($p < 0.05$); ^{A-C} mean within an AS concentration with different letters are significantly different ($p < 0.05$); A means atropine sulfate solution ($\mu\text{g/mL}$). Values are mean $\pm$ standard deviation ($n = 3$).

4 Conclusion

The present findings demonstrate that the crude extract of *T. laurifolia* leaves significantly increased the antioxidant enzyme activities, CAT, and GPx while it decreased the MDA level in the HEK293 kidney and HepG2 liver cells but not in RAW264.7 macrophage cells. The crude extract was shown to have effective healing and protection functions as an anti-insecticide in cell types with different sensitivities. The AS standard drug protected against cell death due to the insecticide toxins, especially in the RAW264.7 cells. This experiment supports the Thai folklore wisdom of using *T. laurifolia* tea as an anti-insecticide agent but further study for more safety details is still needed.

Acknowledgements

The authors would like to thank Faculty of Agro-Industry, Interdisciplinary Graduate School of Nutraceutical and Functional Food, Prince of Songkla University, Culinary Arts and Kitchen Management, Faculty of Hospitality Industry, Dusit Thani College, and Assoc. Prof. Dr. Teerapol Srichana for cells support.

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